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A partial reproduction of STREAMLINE: workload-aware auto-tuning of Apache Flink pipelines on a small Kubernetes cluster

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HIGHLIGHTS

- STREAMLINE's ETL experiments partially reproduced on Flink, Kafka and Kubernetes
- At saturation, Flink's default had 24–32% less throughput than maximum parallelism
- A single-key benchmark made throughput depend on random slot placement
- STREAMLINE's one-node TAXI plan, deployed per vertex, queued like Flink's default
- Fixing the forecaster cut job-wide emulated TAXI cost by 19% over 11 seeds

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ABSTRACT

STREAMLINE auto-tunes Apache Flink pipelines: a Transformer forecasts the workload, neural and polynomial models predict operator behaviour, and NSGA-II chooses each operator's parallelism and buffer size. For its ETL pipeline on a 30-node cluster, the original study reports a 76% latency reduction against Flink's default configuration (FD) on FIT, and cost reductions of 51% and 67% against FD and maximum parallelism (FMAX) on TAXI. We partially reproduced these experiments, without the ContTune baseline and the seamless update, with the authors' unmodified code on a Kubernetes cluster with four one-core Flink TaskManagers. We built the unreleased Flink integration, harness and training data, and measured throughput, end-to-end latency between Kafka brokers and an emulated (proxy) cost over three runs per configuration. At saturation FD processed 24% (FIT) and 32% (TAXI) less than FMAX over all runs; on FIT the gap was 33% when the two busy subtasks ran on different TaskManagers. When peaks exceeded FD's capacity, STREAMLINE's configurations cut median latency by about 99%, confirming the direction of the paper's 76% but not its size. The cost savings were not reproduced. Deployed with their planned per-vertex parallelism, STREAMLINE's TAXI plans reached the paper's one-node cost on our cluster but queued like FD at the peak: every record has the same key, so one subtask does all the work, which STREAMLINE's models cannot represent. A hand-designed control carried the peak at half FMAX's cost. Fixing two forecaster defects cut the expected emulated TAXI cost of job-wide deployment by 19% across 11 seeds.

1. Introduction

Stream processing systems such as Apache Flink [1] run continuously over data whose rate changes by the hour. A configuration fixed at deployment time is either too small for the peaks, so records queue and latency grows, or too large for the troughs, so paid resources sit idle. Choosing operator parallelism, buffer sizes and operator placement by hand does not keep up with such changes, and this is the motivation for runtime auto-tuners [2,3]. Reactive controllers such as Dhalion [4] and DS2 [5] adjust parallelism from observed metrics, predictive ones such as Demeter [6] act on a workload forecast, and Siachamis et al. compared several of them experimentally on the same engine [7]. Our earlier systematic review [8] covered 51 studies on adaptive runtime management in stream processing. Among them, STREAMLINE [9] stood out because its code [10] and benchmark [11] are public, its datasets are public, and it combines workload prediction with configuration tuning and scheduling.

STREAMLINE was published in *Internet of Things* in 2025 [9]. It claims improvements of up to 4× in throughput, 10× in end-to-end latency and 9× in CPU utilisation, and cost reductions of up to 10×, against three baselines: Flink's default configuration (FD), maximum parallelism (FMAX) and the ContTune tuner (CT) [12].

This reproduction had three objectives:

(1) **O1. Decision layer.** Run the released STREAMLINE code, check that it reproduces the decisions documented in its README, and audit its prediction path.

(2) **O2. Big Data experiment.** Reproduce the ETL experiment of the original §5.1.1 (FD vs FMAX vs STREAMLINE on the FIT and TAXI datasets) on a real Flink, Kafka and Kubernetes stack, and compare throughput, end-to-end latency, CPU utilisation and cost with the paper as relative outcomes.

(3) **O3. Extension.** Test whether the defects we found in the released forecaster change STREAMLINE's decisions, and the measured outcome, inside a running cluster.

This is a partial reproduction. The ContTune baseline, the seamless reconfiguration mechanism (Algorithm 5 of [9]), the Yahoo Streaming Benchmark experiments and the multi-pipeline ensembles are out of scope: none of them is in the released code, and rebuilding them would have meant writing new systems rather than reproducing a published one. The paper's comparison with ContTune is therefore not tested. The authors' Beam benchmark exposes only job-wide parallelism on Flink, so we deployed STREAMLINE's per-operator plans in two ways: with one parallelism for the whole job, and with their planned per-vertex parallelism through a Flink cluster option. In neither case was STREAMLINE's placement of individual subtasks enforced (Section 3.5). The STATS, PRED and TRAIN pipelines were left out for time. Section 2 summarises the original study, Section 3 describes our environment and method, Section 4 reports results against the paper's, Section 5 discusses the differences, and Section 6 concludes.

2. Original Study

2.1. Method

STREAMLINE treats a pipeline as a directed graph of operators. For each operator i it chooses two configuration parameters, the parallelism x_i and the network-buffer segment size s_i (16, 32 or 64 KiB), and a node on which to run it. It has four components.

Workload predictor. A Transformer encoder [13] (4 layers, 4 heads, d_model 128) receives the last 100 observations of the source workload and outputs the next 30 (paper) or 50 (released examples) steps. Downstream operator workloads are derived from the source forecast with each operator's input/output ratio.

Performance models. A neural network (two hidden layers of 256 and 128 units) predicts each operator's throughput, latency and CPU use from its input rate, parallelism, segment size and operator type. A polynomial regression of capacity against parallelism gives the throughput an operator can sustain.

Configuration optimiser. NSGA-II [14] (population 20, 20 generations) searches the parallelism and segment-size space for each operator. It trades off throughput, latency and CPU with the weights $\beta = [0.5, 0.25, 0.25]$ and picks the configuration that can carry the forecast peak.

Scheduler and update. A resource-aware scheduler packs operators onto as few nodes as possible. A savepoint-based "seamless" update then applies the new configuration, keeping the old job running until the new one has caught up.

The models are trained offline on profiled runs. Online, STREAMLINE is triggered at deployment and again halfway through each experiment.

2.2. Experimental setup

The authors ran Flink 1.19.1 and Kafka [15] 3.7.0 on a 30-node Kubernetes cluster on Grid'5000 (Gros cluster, Nancy). Nodes were virtualised in three types: type 1 with 4 cores, 8 GB and $0.192/h; type 2 with 8 cores, 16 GB and $0.384/h; and type 3 with 16 cores, 24 GB and $0.768/h. Prices are those of comparable AWS instances. Four RIoTBench pipelines [16] were evaluated, ETL, STATS, PRED and TRAIN, plus YSB. Two datasets were replayed through Kafka:

- **FIT:** MHEALTH body-sensor data [17], a steady workload of about 7,000 elements per second (el/s).
- **TAXI:** New York City taxi trips from 2013 [18], a fluctuating workload of 1,000 to 10,000 el/s.

Throughput was read from the Flink REST API every 5 s, end-to-end latency from Flink latency markers, CPU from per-thread CPU time, and cost from the node prices. Each experiment was repeated three times and averaged.

2.3. Findings relevant to this reproduction

Table 1 lists the paper's statements for the ETL pipeline, which are the targets of our comparison.

Table 1. The original study's ETL results [9, §5.1.1] used as reproduction targets.

Dataset	Statement in the original study
Both	STREAMLINE and FMAX reach the highest throughput, within 1% of each other
FIT	FD throughput 34% lower than STREAMLINE
FIT	STREAMLINE end-to-end latency 76% lower than FD
FIT	STREAMLINE cost 63% lower on average than the baselines; CPU utilisation 211% higher
TAXI	FD handles low load but struggles at higher rates: 14% lower average throughput
TAXI	Average latency of FD and FMAX differs by less than 4%; STREAMLINE 46% lower than CT
TAXI	STREAMLINE cost 51% lower than FD and 67% lower than FMAX, on a single type-2 node
Both	FD has better CPU utilisation than FMAX; FMAX the poorest

3. Reproduction Methodology

3.1. Code and integrity rules

We used two public repositories: STREAMLINE's decision layer [10] (commit d65baec, GPLv3) and the authors' Apache Beam [19] benchmark [11] (commit 57524e2). The benchmark contains the RIoTBench pipelines, the data converter and a Kafka producer. We did not edit either repository. Every fix or alternative behaviour lives in our own scripts, and every difference from the authors' setup is recorded with its reason in a modification log in our repository. The paper's results and ours are kept in separate columns throughout this report.

3.2. Environment

Table 2. Computing environment.

Item	Original study	This reproduction
Hardware	30 virtual nodes on Grid'5000, 10 of each type (280 cores); TaskManagers on 28 of them (248 cores)	One laptop: Intel Core Ultra 7 255HX (20 threads), 31 GB RAM; TaskManagers limited to 4 cores in total
Orchestration	Kubernetes, Flink Kubernetes Operator	Kubernetes via k3d (k3s v1.35) in Docker Desktop (WSL2)
Stream processor	Flink 1.19.1	Flink 1.18.1, the version the benchmark pins (Beam 2.57 FlinkRunner)
Workers	Type 1–3 nodes, slots = cores	4 TaskManagers × (1 CPU, 2 GiB, 2 slots): type 1 scaled by ¼, as in the authors' Helm chart
Messaging, storage	Kafka 3.7.0 (3 brokers in the authors' Helm chart), MongoDB	Kafka 3.7.0 (1 broker, same 1-partition topics), MongoDB 7
Tuner	Python, PyTorch, CUDA	Python 3.8, PyTorch 2.4.1 (CPU), pymoo 0.6.1.2 [20]

The paper's TaskManagers ran on 248 cores and ours on 4, about 1/60 as many. Even the whole laptop, whose 20 hardware threads are shared with Kafka, MongoDB and the operating system, has less than a tenth of the paper's 280 cores. We therefore compare relative outcomes (ratios between configurations), not absolute rates.

Cost model. No cloud resources were rented, so cost is a proxy, not a measured charge. Each one-core TaskManager that hosts at least one task stands for one scaled-down type-1 node and is charged that node's price, $0.192/h; we call this emulated cost. Absolute dollar values are therefore not comparable with the paper's. Because all TaskManagers are identical, cost ratios between configurations equal TaskManager ratios and do not depend on the price chosen.

3.3. Data and preprocessing

FIT. We downloaded UCI MHEALTH [17] (10 subjects, 75.6 MB) and converted it with the authors' `senMlConverter` Docker image into 1.2 million SenML records. **TAXI.** The authors' source, the University of Illinois Data Bank [18], refused our downloads, so we used the Internet Archive mirror of the same 2013 release (checksums verified). Two differences had to be handled. First, the mirror stores taxi and licence identifiers as MD5 hashes, which we mapped to integers by first appearance. Second, its rows are not in time order, so we sorted them by

pickup time. As the converter's README prescribes, we restricted the data to the week of 14–21 January 2013: 3,466,804 trips. A check with the authors' own parser and range filter read 300,000 converted records without an error.

Replay. Records are replayed with the authors' unmodified Kafka producer, which sends each line once its scheduled delay has passed. We found that its thread striding skips every fifth line, so the effective rate is 0.8 times the rate in the file. All rates in this report are effective (measured) rates. We built three kinds of replay schedule, changing only the delay column and never the records: constant rates for capacity measurements, stepped and sinusoidal loads for STREAMLINE's profiling and the FIT evaluation, and the real TAXI arrival pattern, compressed in time by a factor of 220 (the converter's `SCALING` = 1/220).

3.4. Measurement harness

The authors did not release their measurement setup, so we wrote one in Python. For each run it (i) recreates the Kafka topics and restarts the TaskManager pods, because Beam's Kafka reader leaks a polling thread when a job is cancelled; (ii) submits the unmodified benchmark jar with the job-wide parallelism under test; (iii) starts the producer; and (iv) every 5 s records Flink REST metrics per vertex and per subtask, pod CPU, the TaskManager hosting every subtask, and the CPU load of the host itself.

The first 150 s of each run are discarded as JIT warm-up. Runs were interleaved across configurations and repeated three times.

Latency. The paper used Flink latency markers [21]. We found that they mislead in this benchmark for three reasons. Their histograms are long-lived, so warm-up values persist. Each marker goes to one random downstream subtask, so at parallelism p a subtask's 60-marker window spans about $60 \cdot p$ seconds. And markers start at the source operator, so they miss time spent waiting in Kafka, which dominates under overload. We therefore measured **end-to-end latency between the Kafka brokers**. Both topics use LogAppendTime, so every record carries the time at which the broker appended it, and a record's latency is the append time of the i-th output record minus that of the i-th input record. Like the event-time latency of Karimov et al. [22], it includes the time a record waits in Kafka before Flink reads it. It excludes the producer's send path and the consumer's read path, so it is not the full producer-to-consumer latency of an application. An order check on sampled records confirmed the one-to-one matching in every run (match fraction 1.0).

Validity controls. The cluster shares a laptop with other processes, so we logged host load per sample and checked the Windows power log per run. Runs that overlapped CPU-heavy host jobs, battery power or Modern Standby were excluded and re-measured. Battery power alone raised CPU per record by 25–40%. Eighteen runs were re-measured for these reasons, and the excluded runs are kept in separate folders.

3.5. STREAMLINE in the loop

The released code contains the decision layer only, so we connected it to the cluster ourselves.

Operators. Beam fuses the ETL job into two Flink vertices, which we model as two STREAMLINE operators: v_0, the source chain, and v_1, Interpolate through the sinks.

Training data. Twelve profiling runs (parallelism 1, 2, 4, 8 × segment size 16, 32, 64 KiB) under a stepped load gave about 1,000 training rows in STREAMLINE's input format. The authors' collector is unpublished, so the column mapping is ours and documented.

Workload history. The forecaster was trained on 30 minutes of measured source workload.

Hyperparameters. We used the released example's settings (input 100, output 50, 200 epochs) rather than the paper's Table 2 (output 30, 500 epochs), because the decision code expects the example's horizon.

Mapping to Flink. Beam's FlinkRunner accepts only one parallelism for the whole job and cannot pin operators to nodes. A per-operator plan therefore became job parallelism = max over operators, TaskManagers = $\lceil p/2 \rceil$, and the chosen segment size as a TaskManager setting. Each configuration was deployed as a fresh job instead of through the seamless update.

Per-vertex deployment (E4). Flink 1.18 has a cluster option, `pipeline.jobvertex-parallelism-overrides`, with which the JobManager overwrites the parallelism of named job vertices when a job is submitted. We set the option on the JobManager to give v_0 its planned parallelism, submitted the unmodified jar with v_1's parallelism, and set the number of slots per TaskManager so that the job occupied exactly the TaskManagers STREAMLINE planned. The harness checked through the REST API that every vertex ran at the requested parallelism. Flink, not STREAMLINE, still chose the slot of each subtask.

Variants (O3). V0 is the code as released. V4 fixes two forecaster defects: dropout left active at inference, and the input window fed to the network without the scaling used in training. Both load the same trained models and were run with 11 random seeds.

3.6. Experiments

Table 3. Experiments.

ID	Question	Design
E0	Does the released decision layer work, and is it correct?	README examples; code audit; 2^3 factorial correction study (88 runs)
E1	Capacity of FD vs FMAX	p = 1, 2, 4, 8; constant overload; 360 s; 3 reps; FIT and TAXI
L1	Does parallelism buy latency below capacity?	FD, p = 4, FMAX at ~780 el/s; broker-to-broker e2e latency; 3 reps
E3	STREAMLINE V0 and V4 vs FD and FMAX (FIT)	Sinusoidal workload peaking at 1,599 el/s; decision on a 250 s forecast horizon; 3 reps per configuration
E3-T	Same on TAXI's real arrival pattern	Mon–Fri 14:00 history; horizon Fri 14:00 – Sat 05:17 (real time), effective 279–1,752 el/s; 3 reps
E4	STREAMLINE's TAXI plans with per-vertex parallelism	Per-vertex parallelism; V0's plan, V4's plan and a placement-aware control; E3-T workload; 3 reps

4. Results and Comparison

4.1. Decision layer (E0)

The two README examples ran and reproduced the documented decisions. The audit found three defects in the prediction path: (i) in the two-pipeline example, the second pipeline's workload update is written under the wrong key (`v_0` instead of `v_5`); (ii) the forecaster is left in training mode, so dropout is active at inference and forecasts change from call to call; and (iii) the input window is not scaled the way the training data was.

A fourth issue, a checkpoint alias, returns the last epoch rather than the best, but had no measurable effect.

In a 2^3 factorial study on the authors' example workloads, fixing all three defects cut the mean forecast error from 317.9 to 23.4 el/s (−93%) and removed negative forecasts, which appeared in 11.5% of steps before the fix. The fixes interact, so their effects

do not simply add. The released code is runnable, but its forecasts are noisier than the paper implies.

4.2. Capacity: FD vs FMAX (E1)

Table 4. Saturated throughput, mean ± SD over 3 runs (measured in this reproduction). CPU utilisation is the mean CPU use of the TaskManagers that host tasks (Kubernetes metrics API, 5 s samples after warm-up) divided by their 1-CPU limit, so 1.00 means the limit was fully used. Our harness's summary had also counted the idle TaskManagers' background CPU (0.02–0.05 cores each) and gave FD 1.06–1.12; the values here are recomputed from the per-pod samples.

Configuration	TMs	FIT throughput (el/s)	TAXI throughput (el/s)	CPU utilisation, FIT / TAXI	Emulated cost ($/h)
FD (p = 1)	1	1,115 ± 32	1,337 ± 12	1.00 / 1.00	0.192
p = 2	1	1,100 ± 30	1,296 ± 14	1.00 / 0.99	0.192
p = 4	2	1,478 ± 344	1,528 ± 365	0.69 / 0.60	0.384
FMAX (p = 8)	4	1,473 ± 314	1,959 ± 16	0.36 / 0.40	0.768

Every run was saturated: the offered load was 2,620–2,742 el/s and the bottleneck subtask was 100% busy in all 24 runs. FD processed 24% (FIT) and 32% (TAXI) less than FMAX. FD's single TaskManager ran at its CPU limit, while FMAX used 36–40% of its four. The large standard deviations at p = 4 and on FIT at p = 8 do not come from measurement noise. Fig. 1 shows their source.

The ETL job keys every record with one constant key (`WithKeys.of("")`) before its Interpolate step, and the input topic has one partition. So, whatever the parallelism, one source subtask and one Interpolate subtask do all the work. Extra parallelism helps only if Flink's slot assignment puts these two on different TaskManagers, and that assignment varies from run to run. Key skew is a known limit on stream parallelism [23]; here it is total, because there is only one key.

With the two subtasks separated, throughput was 1,630–1,684 el/s on FIT and 1,940–1,969 el/s on TAXI; with them co-located, every configuration fell back to FD's level. The FIT figure of −24% therefore mixes both cases. Counting separated runs only, FD is 33% (FIT) and 32% (TAXI) below FMAX, close to the paper's 34% on FIT. In one FIT run FMAX was co-located and reached 1,111 el/s, the same as FD at four times the cost. Over all 18 runs at p = 8 in this project, 3 were co-located.

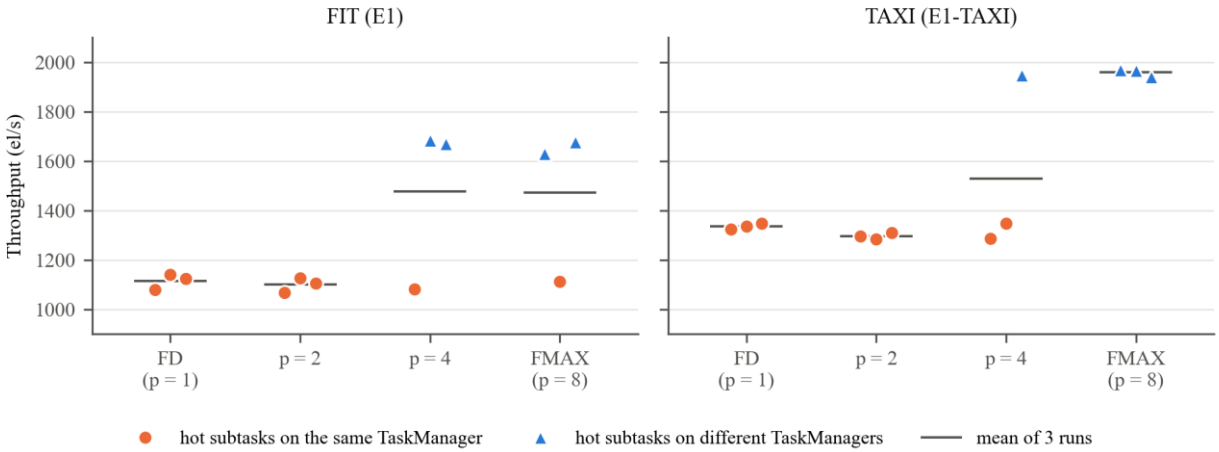


Fig. 1. Saturated throughput of each run. Colour shows whether Flink placed the active source subtask and the single busy Interpolate subtask on the same one-CPU TaskManager.

4.3. Latency below capacity (L1)

At about 780 el/s, below every configuration's capacity, end-to-end latency was the same for FD, p = 4 and FMAX in 8 of 9 runs: p50 33–39 ms, p95 98–115 ms, p99 130–160 ms. FD therefore gives the same latency at a quarter of FMAX's cost. One FD run had a 30 s throughput stall of unknown cause; with only 20% headroom, the backlog took about 90 s to clear. Latency markers, measured in separate runs, reported multi-second p95 values at p = 4 and 8. Those values are an artefact of how markers are routed (Section 3.4).

4.4. STREAMLINE in the loop on FIT (E3)

The FIT evaluation workload is a sinusoid between 201 and 1,599 el/s, so its peaks exceed FD's capacity (about 1,100 el/s). Table 5 and Fig. 2 summarise forecasts, decisions and outcomes.

Table 5. E3 on FIT (measured). Baselines: FD p50 17,228 / 21,839 / 16,804 ms, lower bounds because records still queued at the end of a run were not counted; FMAX p50 62 / 63 / 21,074 ms (the last run co-located).

	V0 (as released)	V4 (corrected)
Forecast error over the horizon (MAE)	115 ± 11 el/s, negative values in 7 of 11 seeds	41 el/s, deterministic
Decision (v_1 parallelism)	7 in 10 of 11 seeds	8 in 11 of 11 seeds
Applied in Flink	p = 7, 4 TMs, 64 KiB	p = 8, 4 TMs, 64 KiB
Emulated cost: STREAMLINE's plan / in Flink	$0.384 / $0.768 per h	$0.384 / $0.768 per h
e2e p50 per run (ms)	117 / 118 / 63	61 / 400 / 66
e2e p99 per run (ms)	1,023 / 3,695 / 397	330 / 4,831 / 1,010

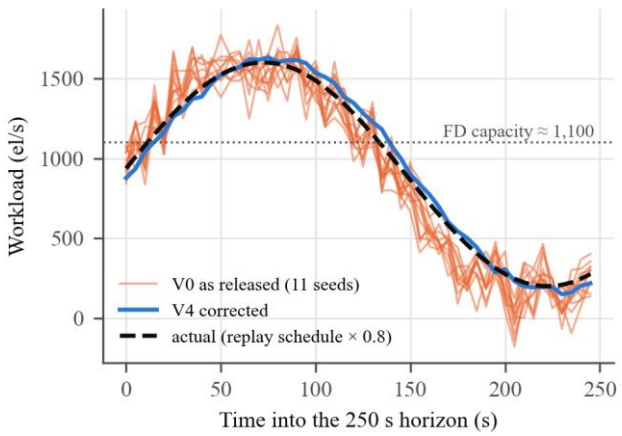


Fig. 2. E3 on FIT: STREAMLINE's 50-step forecasts (V0 for 11 seeds, V4) against the workload that actually arrived.

The fixes cut forecast error by 64% and changed the decision. V0's forecasts overshoot the peak by 5–15%, and a forecast peak above about 1,650 el/s makes the cubic capacity model prefer p = 7 over p = 8. However, both decisions cost the same once mapped to Flink, and their latencies overlap run by run, so the fix changed the decision but not the measured outcome. A seasonal-naive forecast [24] (the value one period earlier) had a lower error (19 el/s) than either variant on this periodic workload.

Against FD, STREAMLINE's configurations reduced median latency by 99% (106–176 ms vs 18,624 ms) and expected p99 by 94–95%. This is not a replication of the paper's −76%; it confirms only the direction. The size is larger because our evaluation workload was designed to overload FD, and below FD's capacity the difference was zero (Section 4.3). Against FMAX they were equivalent. On cost, STREAMLINE's own plan ($0.384/h) would save 50% against FMAX, but in Flink it needs all four TaskManagers ($0.768/h): 300% more than FD and no saving against FMAX.

4.5. STREAMLINE in the loop on TAXI's real workload (E3-T)

The forecaster was trained on the measured ingestion of Monday 00:00 to Friday 14:00, replayed 220 times faster, and the surrogate on 12 TAXI profiling runs. The horizon, Friday 14:00 to Saturday 05:17, carried 279–1,752 el/s (mean 1,278), so its evening peak exceeded FD's capacity (1,337 el/s) but not FMAX's (1,959 el/s). Table 6 and Fig. 3 summarise the results.

Table 6. E3-T on TAXI (measured; 3 runs per configuration). Latency is that of records entering in the horizon, given per run. Throughput is the output rate in the 60 s around the peak.

	FD	STREAMLINE p = 6	STREAMLINE p = 7	FMAX
Chosen by	baseline	V4: 11 of 11 seeds; V0: 3 of 11	V0: 8 of 11 seeds	baseline
TaskManagers; emulated cost ($/h)	1; 0.192	3; 0.576	4; 0.768	4; 0.768
Hot subtasks separated	no (one TM)	3 of 3 runs	3 of 3 runs	1 of 3 runs
e2e p50 (ms)	9,198 / 7,527 / 8,423	55 / 53 / 55	56 / 52 / 54	13,318 / 56 / 9,640
e2e p99 (ms)	20,517 / 17,971 / 18,810	518 / 462 / 315	666 / 177 / 669	24,602 / 397 / 20,340
Throughput at the peak (el/s)	1,375–1,406	1,629–1,630	1,627–1,630	1,351 / 1,629 / 1,366

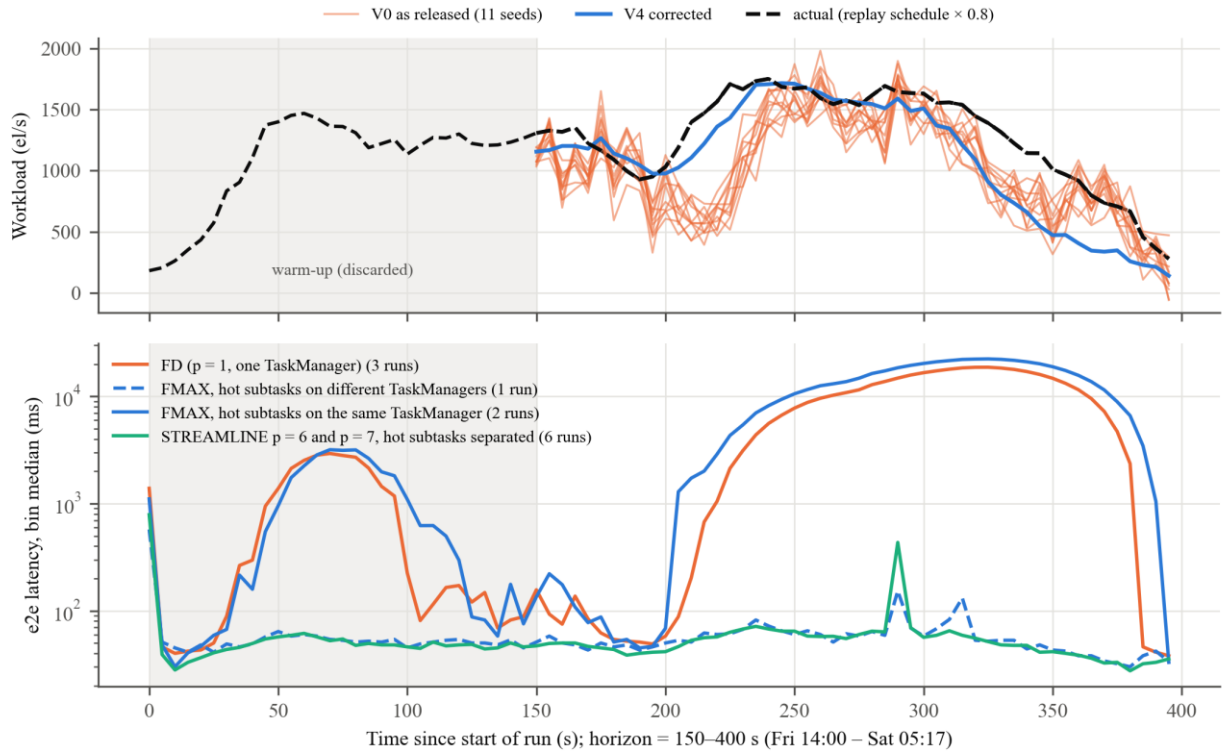


Fig. 3. E3-T. Top: STREAMLINE's forecasts made at the start of the horizon against the workload that arrived. Bottom: median end-to-end latency per 5 s, with runs grouped by configuration and by whether the two busy subtasks shared a TaskManager.

Both variants underestimated the horizon's mean (forecast means 981–1,090 vs 1,278 el/s), but their peaks (1,716–1,829 el/s) matched the actual peak. Fixing the defects cut the error from 297 ± 21 to 210 el/s (MAE). A seasonal-naive forecast, the same hours one day earlier, scored 236 el/s.

Unlike on FIT, the fix changed the cost. V0 chose p = 7 on four TaskManagers in 8 of 11 seeds, V4 chose p = 6 on three in every seed, so the expected emulated deployment cost across the 11 seeds falls by 19% ($0.716 to $0.576/h) at the same latency. All six STREAMLINE runs had their busy subtasks separated and kept the median near 54 ms through the peak, while FD queued for up to 20 s. FMAX again depended on placement: its separated run matched STREAMLINE, and its two co-located runs queued like FD. Because the night-time trough let the queues drain, all but up to four records were emitted by the end of every run, so averaged over the horizon all configurations delivered 1,270 el/s. All unsaturated runs showed a 0.4–1 s latency spike 290–293 s into the run that is not in the input schedule. We could not identify its cause, and it sets most of their p99 values.

STREAMLINE's own plans were cheaper than what Flink could run. V0 packed both operators onto one TaskManager in 8 seeds ($0.192/h), the paper's single-node outcome, and V4 onto two ($0.384/h). Job-wide parallelism with two slots per TaskManager turned these into four and three TaskManagers. Section 4.6 deploys them with per-vertex parallelism.

4.6. Evaluating STREAMLINE's TAXI plans with per-vertex parallelism (E4)

E4 deployed the two plans that STREAMLINE chose in E3-T with per-vertex parallelism (Section 3.5), on the same workload and with the same analysis: V0's plan in 8 of 11 seeds ($v_0 = 1$ and $v_1 = 7$ on one TaskManager) and V4's plan in every seed ($v_0 = 1$ and $v_1 = 6$ on two). As a control we added a workload-specific, hand-designed configuration, which is neither a STREAMLINE decision nor a general scheduling method: $v_1 = 2$ on two TaskManagers with one slot each. The busy Interpolate subtask was the last one in all 100 runs we inspected, and v_0's subtask shares the first slot, so this plan separates the two busy subtasks by construction. Table 7 gives the results.

Table 7. E4 on TAXI (measured; 3 runs per configuration), with FD from E3-T for reference. Latency and throughput defined as in Table 6.

	V0's plan	V4's plan	Control (ours)	FD (E3-T)
Vertex parallelism (v_0, v_1)	1, 7	1, 6	1, 2	1, 1
TaskManagers; emulated cost ($/h)	1; 0.192	2; 0.384	2; 0.384	1; 0.192
Hot subtasks separated	no (one TM)	0 of 3 runs	3 of 3 runs	no (one TM)
e2e p50 (ms)	8,963 / 11,915 / 17,145	8,336 / 10,012 / 14,053	44 / 44 / 46	9,198 / 7,527 / 8,423
e2e p99 (ms)	20,581 / 25,324 / 33,390	18,983 / 21,995 / 29,198	296 / 254 / 474	20,517 / 17,971 / 18,810
Throughput at the peak (el/s)	1,241–1,382	1,290–1,390	1,627–1,631	1,375–1,406

With their planned vertex parallelism and TaskManager count, STREAMLINE's plans cost what it intended: V0's plan as much as FD and 75% less than FMAX, and V4's plan 50% less than FMAX. Neither carried the peak. Both processed about 1,300 el/s at the peak while about 1,630 el/s arrived, like FD, and their median latency rose to 8–17 s. In V0's plan the source and the busy subtask share one one-core TaskManager, which ran at 97–99% of its CPU limit. In V4's plan Flink put the source and the busy subtask on the same TaskManager in all three runs. STREAMLINE's own mapping does the same, since it places v_0's subtask together with subtask 5 of v_1, so enforcing it would not have helped. The control kept up with the peak at a median of 44–46 ms, no higher than STREAMLINE's job-wide configurations in E3-T (52–56 ms), at half FMAX's emulated cost and a third less than V4's job-wide deployment. Under per-vertex deployment the forecaster fix no longer saved money: V4's expected cost across the seeds was $0.384/h against V0's $0.244/h, and both queued. These findings establish a placement-sensitive limitation under our one-core TaskManager environment; they do not establish that the same resource plans would fail on the original multicore cluster.

Other processes on the host used 3.3–6.6 cores during E4, against 1.4–2.4 during E3-T. The slowest V0 and V4 runs coincide with the highest load, but both plans already fell behind at 3.3–3.5 cores, while the control kept up at 4.7–4.9. One V4 attempt failed during task deployment and was repeated.

4.7. Summary against the original study

Table 8. The original study's ETL claims against our measurements.

Claim (ETL)	Original [9]	This reproduction	Verdict
FD throughput vs STREAMLINE/FMAX, FIT	−34%	−24% all runs; −33% with favourable placement	Reproduced (placement-dependent)
FD throughput, TAXI	−14% (time-averaged, fluctuating load)	−32% at saturation; −15% in the 60 s around the TAXI peak; 0% over the whole horizon	Reproduced; size depends on the averaging window
STREAMLINE ≈ FMAX throughput	within 1%	STREAMLINE chose FMAX-like configurations; equal when placement equal	Reproduced
Latency vs FD, FIT	−76%	−99% p50 with peaks above FD capacity; 0% below capacity	Direction only; size not replicated (FD overloaded by design)
FD vs FMAX latency, TAXI	< 4% difference	−8% (mean p50 of 3 runs); per run FD-like when co-located, −99% when separated	Not reproduced quantitatively; strongly placement-dependent
Emulated cost vs baselines	−51% to −67%	Job-wide (E3, E3-T): FIT 0% vs FMAX, +300% vs FD; TAXI (V4) −25% vs FMAX, +200% vs FD. Per-vertex (E4): TAXI V0 plan 0% vs FD and −75% vs FMAX, V4 plan −50% vs FMAX, but both queued like FD at the peak	Not reproduced: the planned cost is reached only by losing the peak
CPU utilisation: FD better than FMAX	qualitative	1.00 vs 0.36–0.40	Reproduced

5. Discussion

5.1. What was reproduced and why

The qualitative structure of the original §5.1.1 holds on our cluster. FD runs out of capacity first, FMAX buys throughput at a high resource price, FD uses its CPU better, and a tuner that sizes the job for the forecast peak avoids FD's queueing. On FIT, the throughput gap between FD and FMAX with separated hot subtasks (−33%) almost matches the paper's −34%. That agreement is partly coincidence, because the two clusters differ in node size, broker count and Flink version. The latency gap depends almost entirely on how far the load exceeds FD's

capacity: zero below capacity (L1), and tens of seconds of queueing when peaks exceed it (E3). The paper's −76% therefore says as much about the workload as about the tuner, and a percentage without the load relative to capacity cannot be compared across clusters.

5.2. Why the cost savings did not reproduce

STREAMLINE's savings come from per-operator parallelism and from packing operators onto fewer, larger nodes. The paper's ETL plan on TAXI ran on a single 8-core node, and on our TAXI data STREAMLINE's own plan was the same in 8 of 11 seeds: one TaskManager. Beam's FlinkRunner sets one parallelism for the whole job, so in E3 and E3-T such a plan needed three or four TaskManagers. A Flink cluster option removed that obstacle without changing the benchmark (E4). The plans then ran at the cost STREAMLINE intended, but they could not carry the TAXI peak, so the cause lies in the models rather than in the deployment. Because every record has the same key, one v_1 subtask does all the work, and $v_1 = 7$ has no more capacity than $v_1 = 1$ on the same CPU. STREAMLINE learned a high capacity for high parallelism from profiling runs that spread the job over $\lceil p/2 \rceil$ one-core TaskManagers and usually separated the busy subtasks. Its features (input rate, parallelism, segment size, operator type) cannot show that packing the same parallelism onto one core removes that capacity, or which subtask is busy. Our hand-designed two-TaskManager control shows that a 50% saving against FMAX at equal latency exists on this cluster. Finding it, however, needed knowledge of the key distribution and of Flink's slot assignment that STREAMLINE's models do not have, and the original study does not discuss this case.

5.3. The benchmark limits what tuning can achieve

The single key and single partition mean that ETL's capacity is set by two subtasks and by whether they share a CPU, an event that Flink's scheduler decides at random. In our data this made the difference between 1,100 and 1,680 el/s on FIT and between 1,320 and 1,960 el/s on TAXI. It also leaked into STREAMLINE's training data: two of the six multi-TaskManager TAXI profiling runs were co-located. In an earlier, discarded profiling set, two of three $p = 4$ runs happened to be co-located, which lowered the learned $p = 4$ capacity from 1,503 to 859 el/s and changed the decisions. STREAMLINE's features (input rate, parallelism, segment size, operator type) cannot represent this effect. The original study reports averages of three runs without placement information, so we cannot tell how often its own runs were affected.

5.4. The forecaster

The released forecaster has real defects, and fixing them cut its error by 64–93%. In the running system the effect was on cost, not latency: none on FIT, and on TAXI one TaskManager fewer in most seeds, a 19% lower expected emulated cost of job-wide deployment across 11 seeds. With per-vertex parallelism, both variants' plans queued, and V4's cost more (Section 4.6). Decisions are sensitive to the forecast only near a point where the capacity model's ranking of configurations flips. On FIT that point was itself an artefact of fitting a cubic to four measured parallelism levels: the fitted curve falls from $p = 7$ to $p = 8$ although no $p = 7$ run was ever profiled. A seasonal-naive baseline beat the Transformer on our periodic FIT workload. On TAXI, the corrected Transformer (210 el/s) beat a seasonal-naive forecast from the previous day (236 el/s), and the released version (297 el/s) did not. The original study does not compare its forecaster with simple baselines, and we think it should.

5.5. Reproducibility of the artefacts

The code made a reproduction possible, but several missing or under-specified pieces cost most of our effort, a pattern also reported for computer systems research in general [25]. The Flink integration, the measurement harness, the training data and the rule that turns metrics into training rows were not released. The TAXI and FIT replay scaling ("as recommended") is not specified, and neither is whether latency was measured with markers or end to end. The artefacts also contain defects: the producer silently drops 20% of the data, the benchmark's Prometheus configuration does not work outside the Flink operator because Flink 1.18 keeps the quotes around the port, and the released hyperparameters differ from the paper's Table 2. Finally, the paper ran Flink 1.19.1, but the benchmark pins 1.18. All are documented in our modification log.

5.6. Limitations and threats to validity

Scale. Our TaskManagers had 4 cores against the paper's 248, and Kafka had one broker. Absolute rates are not comparable, and the producer could not offer more than about 2,700 el/s without disturbing the measurement.

Shared host. The cluster shares a laptop with other software. We recorded host load and power state for every run and excluded runs that were disturbed, but small variations remain: other processes used 3–5 cores in most experiments, 1.4–2.4 in E3-T and up to 6.6 in E4 (Section 4.6).

Sample size. We made three runs per configuration, and co-location is a roughly 20% event. Several differences in Section 4 are within run-to-run variation, and we say so where they are. Following Hoefler and Belli [26], we report per-run values (Fig. 1, Table 5) wherever runs disagree, not only means.

Workloads. Capacity runs used constant-rate replays. The FIT STREAMLINE evaluation used a synthetic sinusoid, and only E3-T used real arrival times.

Mapping and cost. STREAMLINE's decisions were applied through job-wide (E3, E3-T) or per-vertex (E4) parallelism and full redeployment, not through the seamless update, and Flink rather than STREAMLINE chose the slot of every subtask. Cost is a proxy derived from TaskManager counts (Section 3.2), not a measured cloud charge.

6. Conclusion

We partially reproduced the ETL experiments of STREAMLINE's §5.1.1 on both datasets with the authors' unmodified code, on a real Flink, Kafka and Kubernetes stack, without the ContTune baseline or the seamless update. The parts the authors did not release (harness, Flink integration, training data) were built by us and documented. The paper's throughput and latency findings were reproduced in direction. On FIT the throughput gap was also reproduced in size when the busy subtasks were placed apart (−33% against the paper's −34%; −24% over all runs), and sizing for the forecast peak removed FD's queueing. The latency reduction was larger than the paper's because our FD was overloaded by design, so its size is not a replication. The same held on TAXI's real arrival pattern, where fixing the forecaster's defects also lowered the expected emulated cost of job-wide deployment by 19% across 11 seeds. The paper's cost savings were not reproduced. Deployed with their planned per-vertex parallelism, STREAMLINE's TAXI plans reached the paper's one-node cost but queued like FD at the peak on our one-core TaskManagers, because its models cannot see that one subtask does all the work. A hand-designed control carried the peak at half FMAX's cost, so the saving exists on our cluster, but STREAMLINE did not find it.

Three lessons go beyond this paper. First, a benchmark's data distribution can cap what any tuner achieves: here a single key turned parallelism into a placement lottery. Second, measurement definitions matter as much as the system: Flink's latency markers hid queueing in Kafka and inflated tails at high parallelism, so latency had to be measured between the Kafka brokers instead. Third, a reproduction on shared hardware needs explicit controls for host load and power state; both changed our results by 20–40% before we controlled them.

CRediT authorship contribution statement

Hamza Alqaruoti: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Project administration (measurement harness, experiments and integration). **Mohammad Alqaryouti:** Software, Resources, Investigation (Big Data framework setup and metrics collection). **Noura Alrahbi:** Methodology, Software, Formal analysis (STREAMLINE audit, correction study and training). **Sulaiman Shawabkeh:** Data curation, Formal analysis, Visualization (datasets, workloads, analysis and comparison with the original study). **All authors:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic), including Claude Opus 5.5 accessed through Claude Code, in order to set up and debug the cluster environment, write and review the measurement, data-preparation and analysis scripts, read and audit the original authors' code, and draft and edit the documentation and this report. After using this tool, the authors reviewed and edited the content as needed, checked every reported number against the raw result files in the repository, and take full responsibility for the content of this report.

Data availability

All code, configuration, raw results and analysis scripts are available at https://github.com/HamzaOsama1994/streamline-reproduction. This includes per-sample metrics, per-run placement logs and complete per-record latency traces for E3-T and E4. The original authors' repositories [10,11] are used unmodified at the commits given in Section 3.1. The datasets are public: UCI MHEALTH [17] and the 2013 New York City taxi data [18].

Appendix A. Division of work

Table A.1. Division of work.

Team member	Assigned responsibility	Tasks completed	Individual outputs
Hamza Alqaruoti Team lead	Experiments, measurement and integration	Measurement harness Profiling, E1, L1, E3, E3-T and E4 runs Validity controls (host load, power state) Report integration	Harness and raw results Section 3.4; final report
Mohammad Alqaryouti	Big Data framework setup	k3d cluster with Flink, Kafka and MongoDB Benchmark build Metrics collection	Kubernetes manifests Environment record (Section 3.2)
Noura Alrahbi	Method implementation	STREAMLINE audit and correction study Training on profiling data Mapping to Flink	Integration scripts Sections 3.5 and 4.1
Sulaiman Shawabkeh	Datasets, workload and analysis	FIT and TAXI preparation Replay schedules Statistics, figures and comparison with the original study	Data and analysis scripts Sections 3.3, 4 and 5

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